



AWS CodePipeline

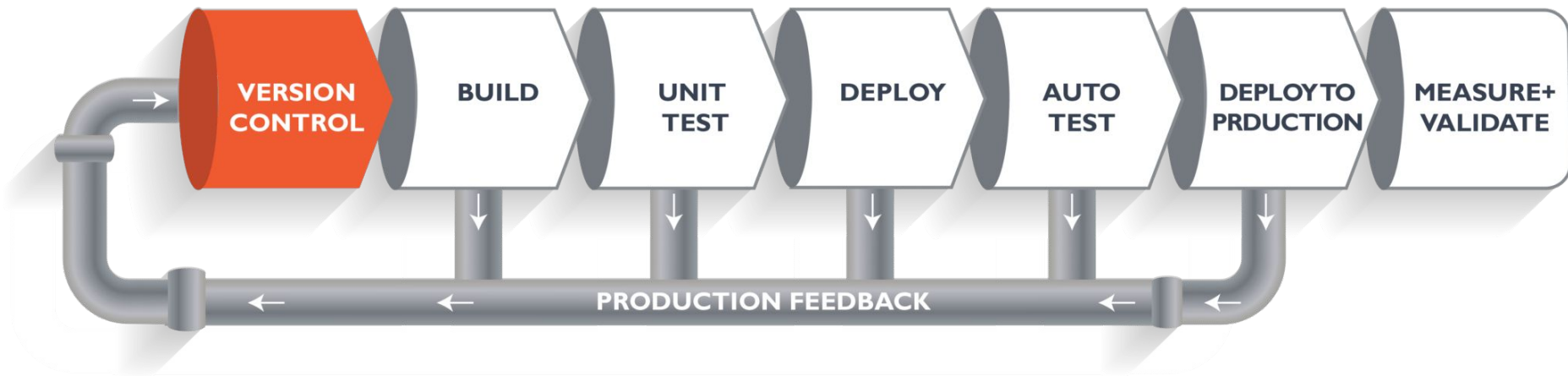




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CI/CD

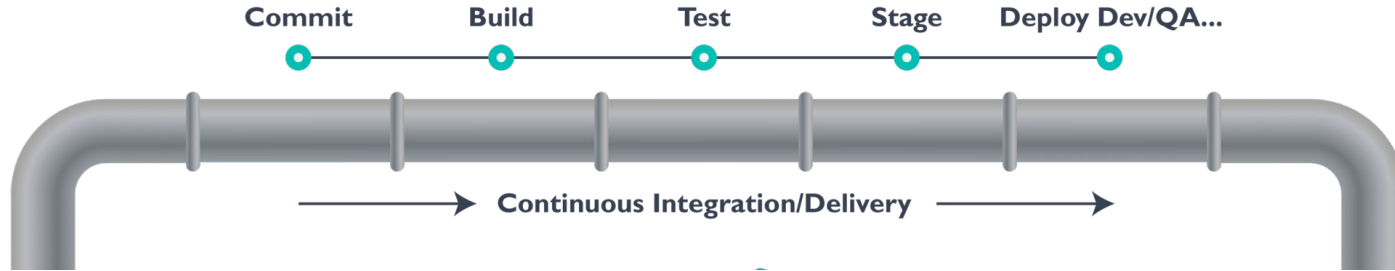




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CI/CD Pipeline Stages and Open Source Solutions

CI/CD Pipeline



Gitlab



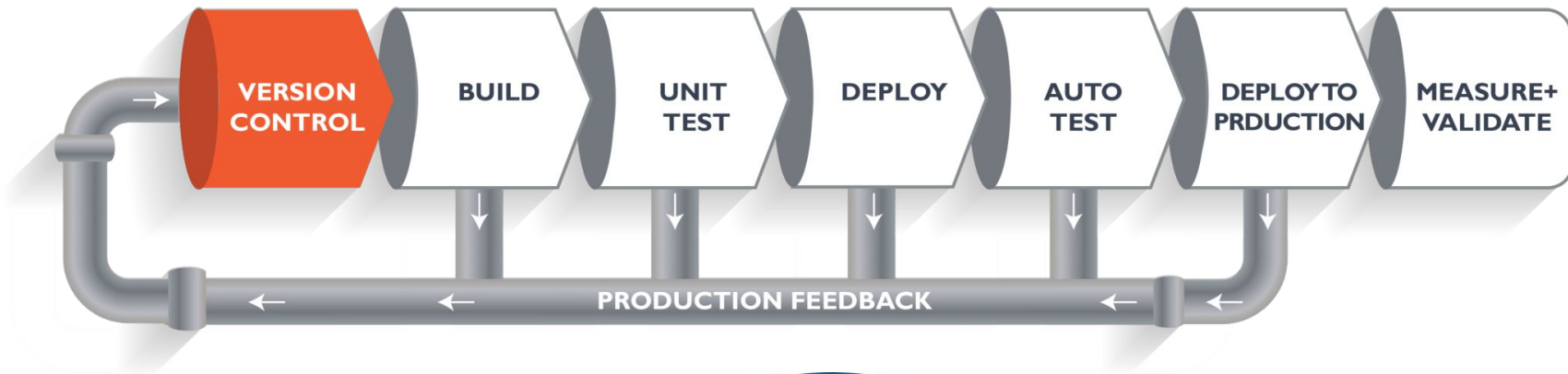
Teamcity



Version Control Stage



Version control, also known as source control, is the practice of tracking and managing changes to software code. Version control systems are software tools that help software teams manage changes to source code over time.



Bitbucket
Atlassian



Gitlab
Gitlab

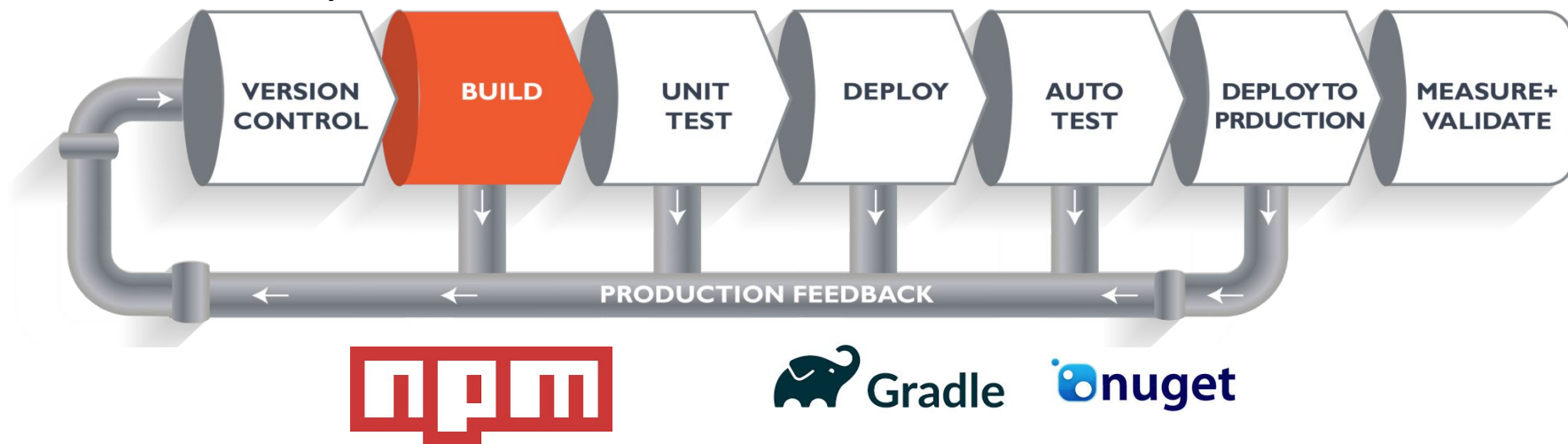
Build Stage



Build stage includes compiling and linking source code pushed within version control systems (like github) to create an executable file.

For eg:-

- .exe - An executable program that can be run on Microsoft Windows operating systems. Examples of executable files on Windows include: Word.exe, chrome.exe etc.
- .rpm - Redhat linux installer package
- .bin - Linux binary executables

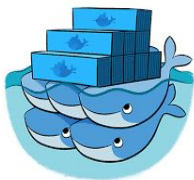


Deploy Stage



The deploy stage is the initial part of the deployment process where the application is released into the staging(pre-prod) environment, making it accessible to developers for testing and validating.

This initial deploy stage helps the developers to analyze and understand whether the application is prepared for the production environment and is operating as planned in the non production environment



kubernetes



ANSIBLE

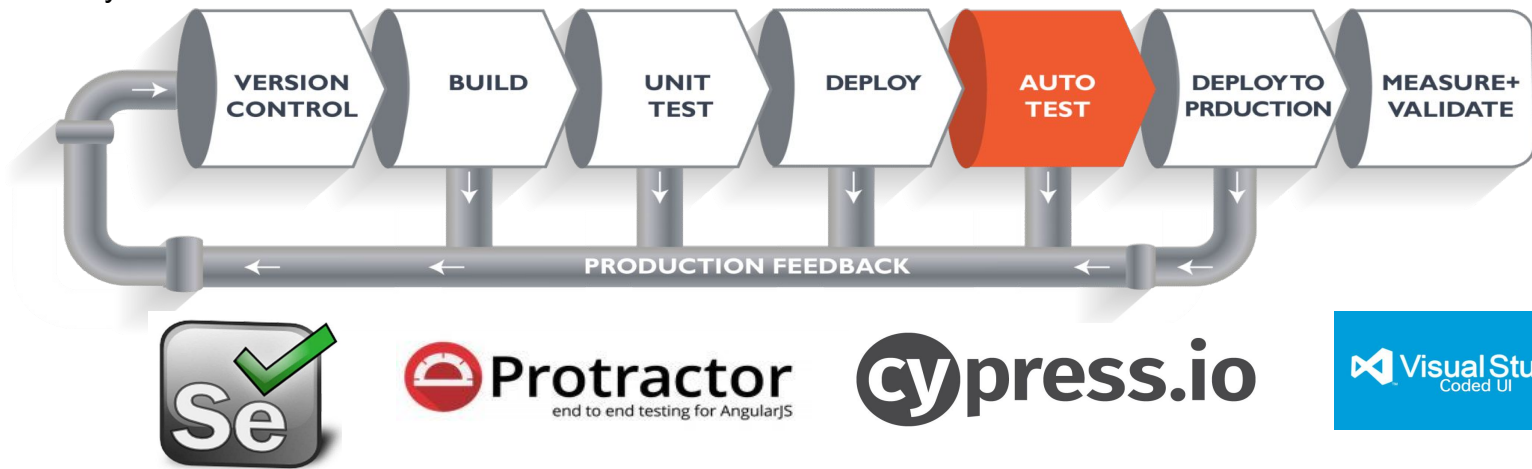


Auto Test Stage

CI CD testing is one of the phases in this process that takes care of quality assurance and perfect delivery of the software with zero errors. The goal of CI/CD testing is to promote the release only when it passes the functional requirements testing, remains stable, and is free of defects. Typically, it is an automated process that is fully integrated with the CI CD pipeline.

Common testing use cases are,

- User Acceptance Tests
- Functional Tests
- Load Tests
- Security Tests



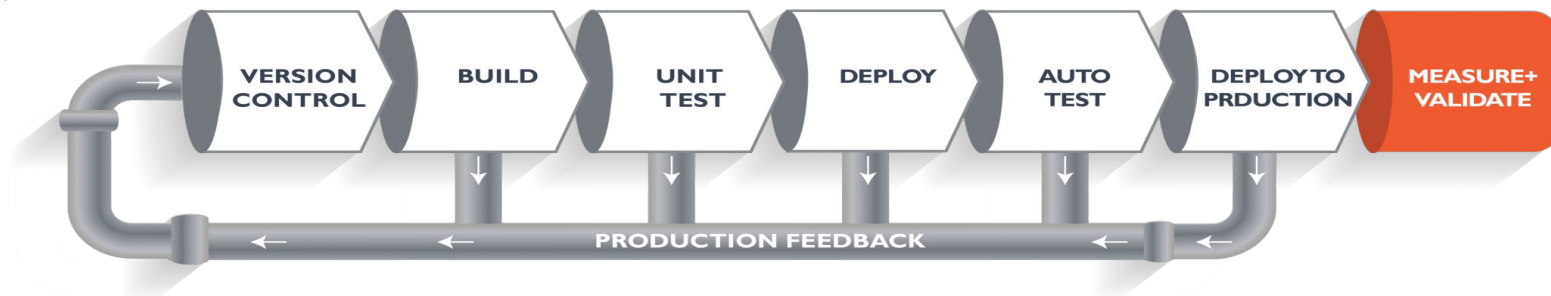


Measure + Validate

Continuous improvement involves collecting and analyzing feedback on what you've built or how you're working in order to understand what is performing well and what could be improved.

Having applied those insights, you collect further feedback to see if the changes you made moved the needle in the right direction, and then continue to adjust as needed.

This stage is considered as “Observability” stage where we observe everything of the deployed application including logs, usage patterns, attacks, outages etc and use tools like prometheus, grafana, kibana for logging and visualization.



Nagios[®]



Continuous Delivery vs Continuous Deployment

Continuous Integration



Continuous Delivery



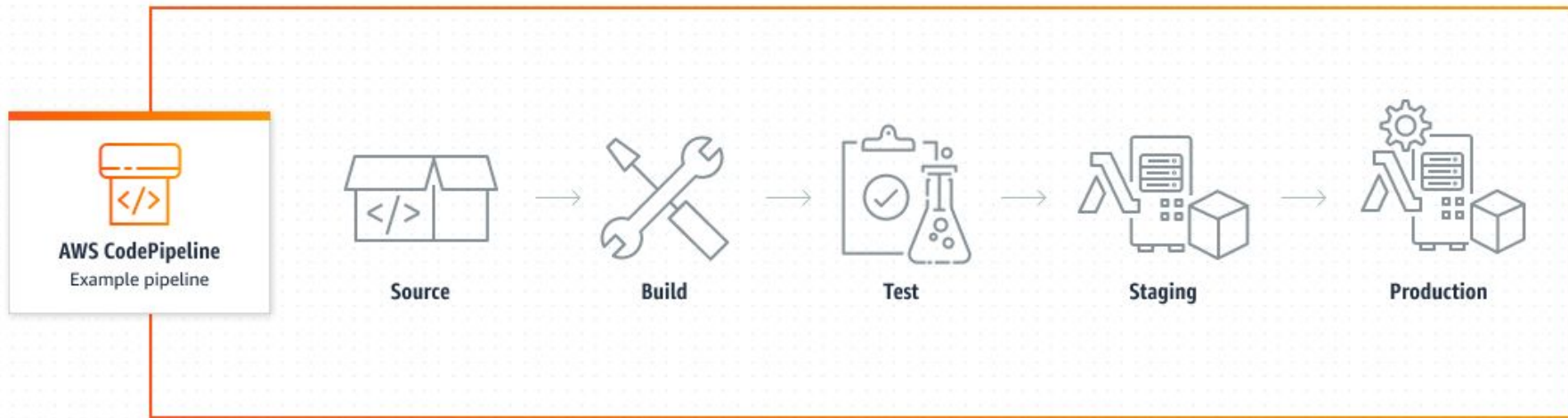
Continuous Deployment





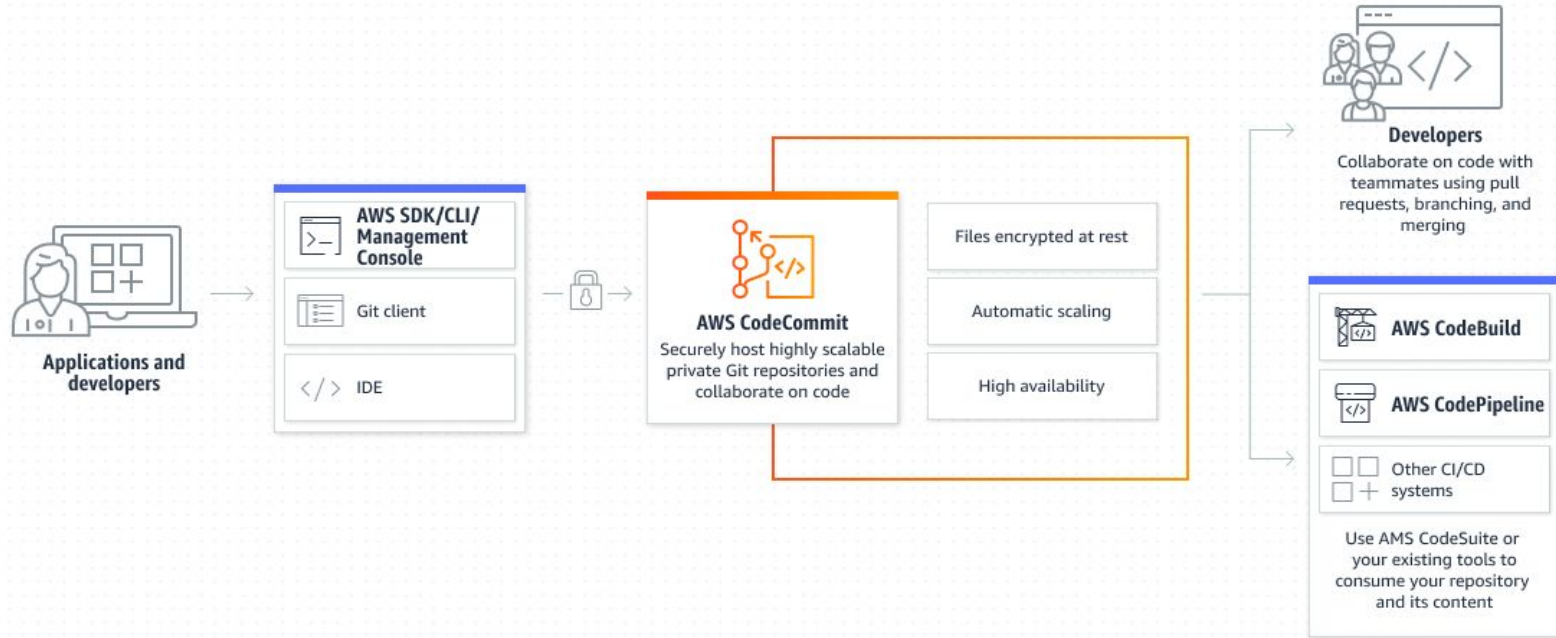
CI/CD on AWS

AWS CodePipeline



AWS CodePipeline is a continuous delivery service you can use to **model, visualize, and automate the steps required to release your software.**

AWS CodeCommit



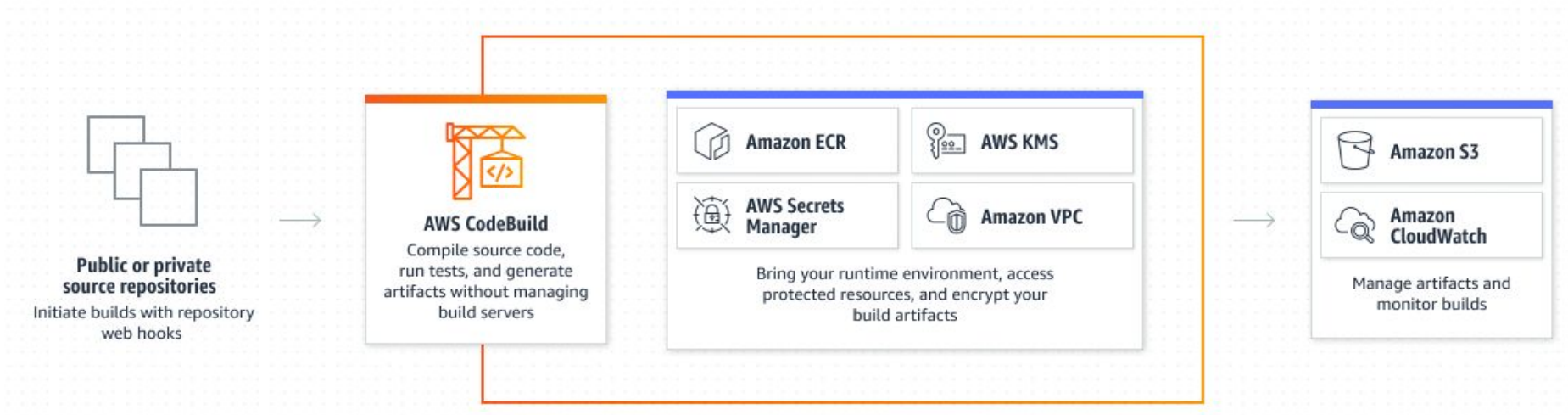
AWS CodeCommit is a **version control service** hosted by Amazon Web Services that you can use to **privately** store and manage assets (such as documents, source code, and binary files) in the cloud.



AWS CodeCommit

- It supports the standard functionality of Git, so **it works seamlessly** with your **existing Git-based tools**.
- It hosts **private repos**
- CodeCommit repositories are **encrypted at rest as well as in transit**.
- **Support pull requests** where users can review and comment on each other's code changes before merging them to branches
- CodeCommit has **no limit on the size** of your repositories
- You can **migrate to CodeCommit from any Git-based repository**.

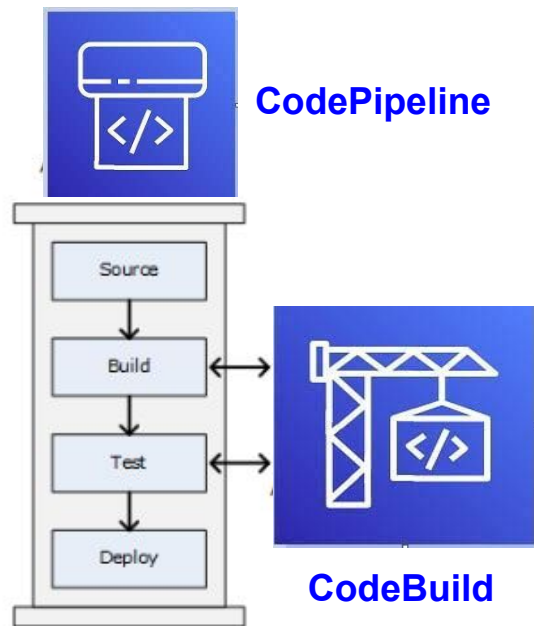
AWS CodeBuild



- AWS CodeBuild

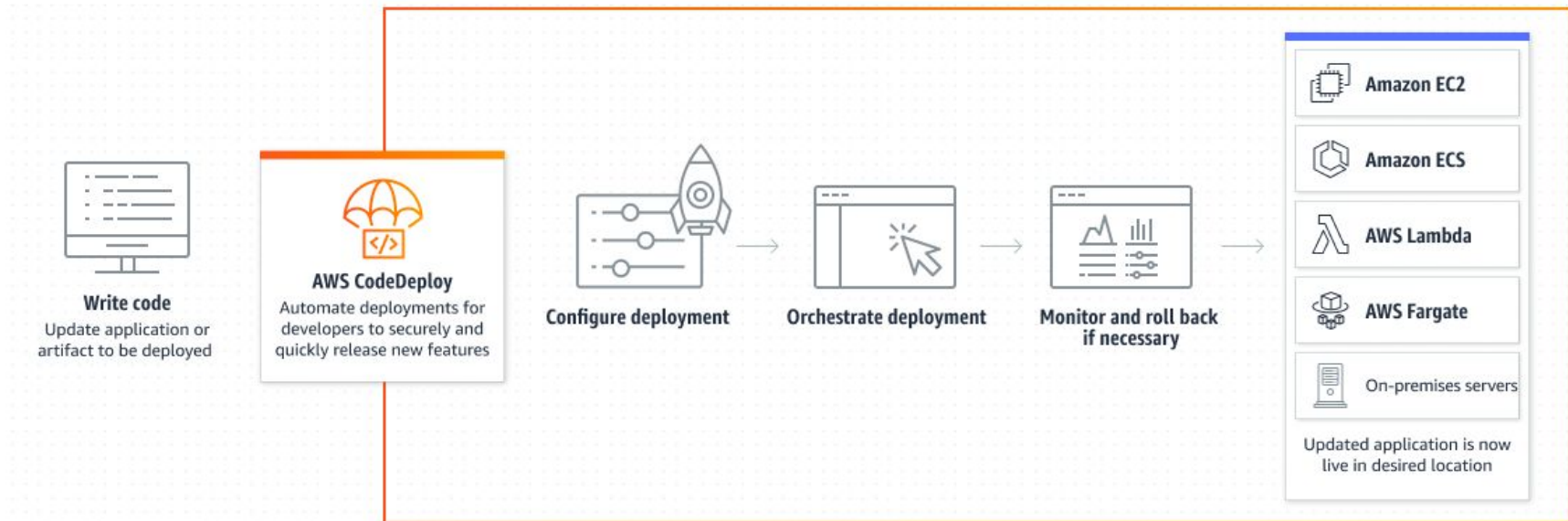
- **compiles** your source code,
- **runs unit tests**,
- **produces artifacts** that are ready to deploy.

AWS CodeBuild



- **Fully managed** – CodeBuild eliminates the need to set up, patch, update, and manage your own build servers.
- **On demand** – CodeBuild scales on demand to meet your build needs. **You pay only for the number of build minutes you consume.**
- **Out of the box** – CodeBuild provides preconfigured build environments for the most popular programming languages. All you need to do is point to your build script to start your first build.
- **Easily reachable** from Codepipeline Console

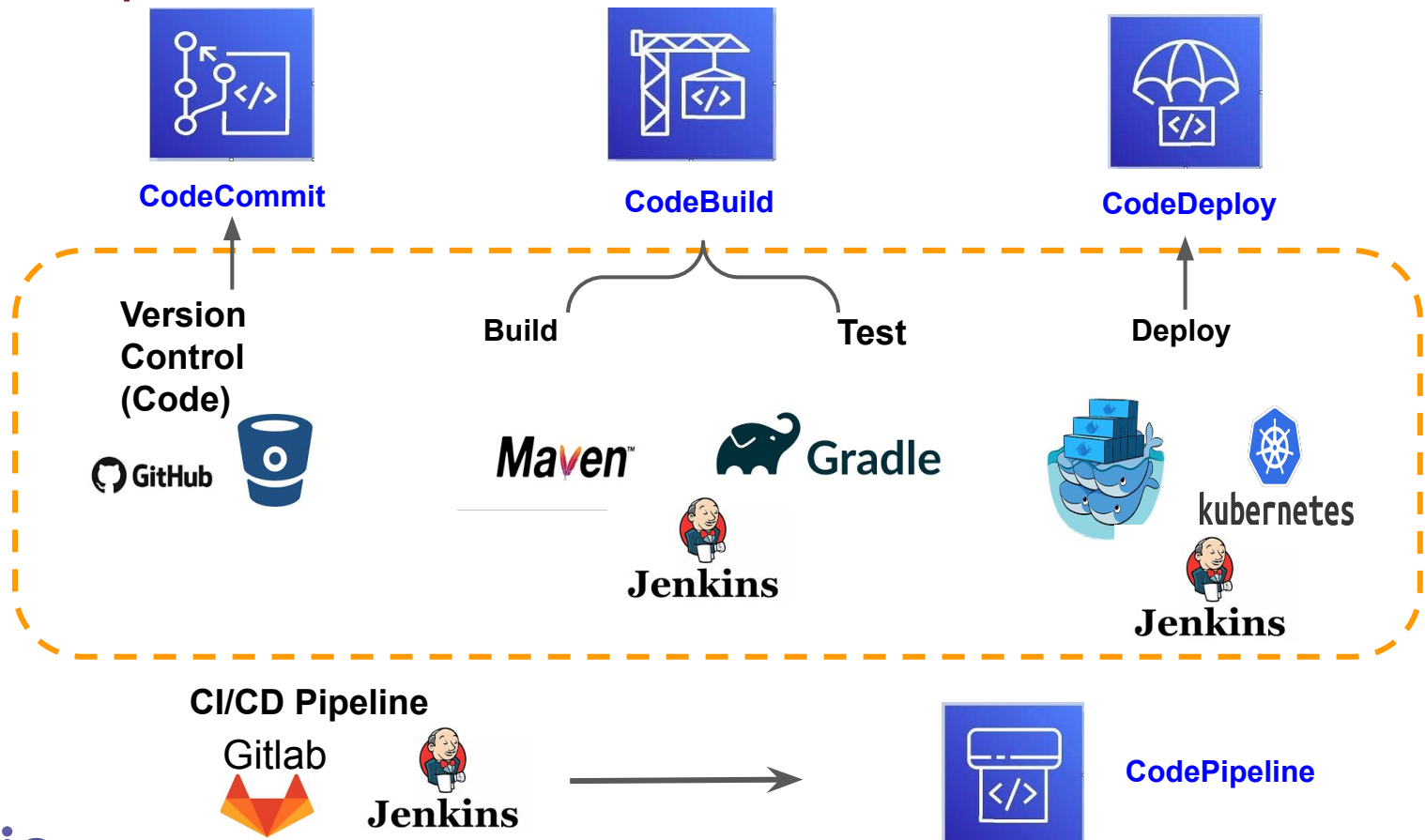
AWS Deploy



AWS CodeDeploy is a fully managed deployment service that automates software deployments to various compute services, such as **Amazon Elastic Compute Cloud (EC2), **Amazon Elastic Container Service (ECS)**, **AWS Lambda**, and **your on-premises servers**.**



AWS Pipeline Structure



THANKS!

Any questions?



AWS CodePipeline

